

24. Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(PRICE)$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

We modify the supply equation by adding *MIXED*:

$$\ln(QUAN_t) = \beta_1 + \beta_2 \ln(PRICE_t) + \beta_3 STORMY_t + \beta_4 MIXED_t + e_{st}$$

The corresponding reduced-form equation for $\ln(PRICE)$ becomes:

$$\begin{aligned} \ln(PRICE_t) = & \pi_0 + \pi_1 MON_t + \pi_2 TUE_t + \pi_3 WED_t + \pi_4 THU_t + \pi_5 STORMY_t \\ & + \pi_6 MIXED_t + v_{t2} \end{aligned}$$

We estimate this reduced-form equation using OLS.

The coefficient on *MIXED* is statistically significant at the 5% level since its p-value is 0.00321. To test the joint significance of *STORMY* and *MIXED*, we perform an F-test. The F-statistic is 16.14 with a p-value of 7.857×10^{-7} . Since the F-statistic is greater than 10, we conclude that *STORMY* and *MIXED* are jointly highly significant and the instruments are strong.

b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

We estimate the demand equation using *STORMY* and *MIXED* as instrumental variables for $\ln(PRICE)$ via two-stage least squares (2SLS).

In the first stage, $\ln(PRICE)$ is regressed on *MON*, *TUE*, *WED*, *THU*, *STORMY*, and *MIXED* to obtain fitted values. In the second stage, $\ln(QUAN)$ is regressed on price_hat and the weekday dummies.

The estimated coefficient on price_hat is -0.9301 and is statistically significant at the 1% level ($p = 0.0095$), indicating a negative price elasticity of demand. Compared to the OLS results in Table 11.5, where the price elasticity is -1.1194, the IV estimate is smaller in absolute value (less negative). This suggests that the OLS estimate is biased downward in this case.

The IV results remain consistent with demand theory, showing a negative and statistically significant relationship between price and quantity demanded. The results confirm that price is endogenous, and OLS estimates are biased due to simultaneous determination of price and quantity in the market.

c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

We perform the Sargan test to examine the validity of the instruments *STORMY* and *MIXED*. The null hypothesis is that the instruments are valid.

The test statistic is 0.772 with a p-value of 0.380. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, *STORMY* and *MIXED* are valid instruments, and the overidentifying restrictions are satisfied.

d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

We test the joint significance of *MON*, *TUE*, *WED*, and *THU* in the reduced-form equation for $\ln(PRICE)$. The null hypothesis is that all weekday dummy variables are jointly equal to zero.

The F-statistic is 0.6279 with a p-value of 0.6437. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, the weekday dummy variables are not jointly significant in explaining $\ln(PRICE)$.

Although the weekday dummy variables are jointly insignificant in the reduced-form equation, this does not affect the validity of the instrumental variable approach. Since the instruments (*STORMY* and *MIXED*) are strong ($F = 16.14$) and valid (Sargan $p = 0.380$), we can estimate the supply equation using 2SLS with confidence.